MYP4 Unit 2 – New Formative Assessment (Version B)

1. Which fundamental force is responsible for holding quarks together inside protons and neutrons?

a) Electromagnetic

b) Strong nuclear

c) Weak nuclear

d) Gravitational

2. Which force acts over cosmic distances?

a) Strong

b) Weak

c) Electromagnetic

d) Gravitational

3. Which subatomic particles determine the identity of an element?

a) Electrons

b) Protons

c) Neutrons

d) Both protons and neutrons

4. Give the SI unit used for mass:

5. Give the SI unit used for force:

Section 2: Atomic Structure & Ions (New Situations)

1. A neutral atom of an element contains:27 protons, 32 neutrons, 30 electrons

a. **Describe** what you know about its: mass number, charge, cation/anion, atomic number and deduce which atom this is.

b. Write the atomic notation (A Z X format).

2. A chlorine atom gains one electron.

a. **State** what type of ion is formed (name + charge)?

b. **Explain** the difference between cation and anion.

c. **Explain** what the difference is between **losing** 1 electron or **gaining** 1 proton.

3. Two atoms have the same number of protons but different numbers of neutrons.

a. **State** what are these atoms called.

b. **Outline** one scientific consequence of differing neutron numbers.

4. The relative atomic mass of magnesium is 24.31.

Explain what the decimal value tells us about magnesium atoms.

5. A person travels from Earth to a small asteroid where g = 1.7 N/kg.

The person’s mass is 58 kg.

a. **Calculate** their weight on the asteroid.

b. **Explain** why their mass does not change.

Section 3: Periodic Table Reasoning

1. Complete the table for the given isotopes:

Isotope	Protons	Neutrons	Electrons	Mass number
Silicon - 30				
Boron - 11				

2. Draw electron configuration **diagrams (**using shells) AND normal notation (eg. 2,8,8,3) for:

a. Aluminium (Al)

b. Oxygen (O)

3. **Explain** why atoms are electrically neutral even though they contain charged particles.

4. **Describe** what changes for the atom Oxygen when you add one additional proton.

Section 4: Moles, Atoms & Forces (Fully New Problems)

1. A sample contains 45 g of copper (Cu).

a. Calculate the number of moles of copper.

b. Determine the number of atoms in the sample.

2. A chemist prepares 4 moles of carbon dioxide (CO_2).

a. Calculate the mass of that amount of CO_2.

b. Determine the number of oxygen atoms in 4 moles of CO_2.

3. Two bags of sand are placed 4.2 m apart.

Bag A mass: 85 kg

Bag B mass: 120 kg

Calculate the gravitational force between them.

4. A 1,200 kg satellite orbits a planet where g = 3.9 N/kg.

Calculate the gravitational force acting on the satellite.

5. **Provide** an example of each of these fundamental forces:

a. Strong nuclear force:

b. Electromagnetic force:

c. Weak nuclear force:

d. Gravitational force:

Section 5: Inquiry & Application

In your own words: **explain why** the gravitational force on top of mount everest is lower than at sea level

Predict whether a person standing on Jupiter's moon Europa (0.008 times the mass of earth) would experience more or less weight than on Earth. Justify fully **using a formula**.

Extra spicy iconic overly complex questions:

How many are in ALL of the oceans of the world?

- **Hints:**
  - Water molecule H_2O, what is its molar mass?
  - Amount of grams of H_2O in all the oceans of the world = $1.332*10^{24}$g

Calculate the F_g:

The AICS (m_1=$1.28*10^5$kg) was ripped off the ground by someone very strong (me) and shot into outer space. It is experiencing the gravitational force of the sun (m_2=$1.989*10^{30}$kg). AICS is at a distance of 60 million big macs (8.8cm ech) from the sun's <u>surface</u>. Calculate the F_g!